# Mechanisms · Pupil Resources

Learning goal: I can identify levers, pulleys, and gears and explain how they let a small effort move a heavy load.

## Arrival Questions (In your book)

1. Which box uses a lever and pivot: Box 1 or Box 2?
2. What is the heavy object we want to lift called?
3. What is the push or pull we put in called?
4. Can you ask an adult to help?

## Starter Question

How does a long lever with the pivot close to the load help you lift a heavy box?

A. Your hand pushes a smaller effort force through a longer distance to lift the heavy load a short distance.

B. It makes the box lose all its mass.

C. It works without any push at all.

## Method Strip

1. Place the eraser pivot on the desk and rest the 30 cm ruler across it.
2. Place the load at the 2 cm mark close to the pivot. STOP AND CHECK.
3. Push gently at the 28 cm mark (far from the pivot) and notice how easy the lift feels.
4. Mesh the two card gears together and turn Gear A

clockwise to observe Gear B.

5. Record your lever, pulley, and gear observations in your book.

## Exit Ticket

Name the fixed point that a lever turns around.

Where should the pivot be placed to make lifting a heavy load easier?

How do levers, pulleys, and gears allow a smaller force to have a greater effect?